

A Novel Ambipolar Ferroelectric Tunnel FinFET based Content Addressable Memory with Ultra-low Hardware Cost and High Energy Efficiency for Machine Learning

Jin Luo^{1#}, Weikai Xu^{1#}, Boyi Fu¹, Zheru Yu¹, Mengxuan Yang¹, Yiqing Li¹, Qianqian Huang^{1,2*}, and Ru Huang^{1,2*}

¹School of Integrated Circuits, Peking University, Beijing 100871, China. ²National Key Laboratory of Science and Technology on Micro/Nano Fabrication, Beijing 100871, China. [#]Equal contribution; (*Email: ruhuang@pku.edu.cn; hq@pku.edu.cn)

Abstract

In this work, a novel ferroelectric tunnel FET (FeTFET) based content addressable memory (CAM) cell with only one transistor is proposed and experimentally demonstrated based on 14-nm FinFET technology node for the first time. By exploiting and modulating the non-volatile ferroelectric polarization for entry storage and the unique feature of ambipolar tunneling current for input searching query, XNOR-like matching operation of ternary CAM can be realized in one Fe-FinTFET without the need of twin complementary circuit branches. Moreover, benefiting from the ferroelectric multi-domain feature and the intrinsic steep slope from tunneling mechanism, multi-bit CAM function for high density can also be experimentally implemented in the fabricated single Fe-FinTFET device. Based on the proposed FeTFET CAM design, Hamming and Manhattan distance computing are demonstrated with high energy efficiency, showing its great potential for area- and energy-efficient machine learning.

Introduction

Content addressable memory (CAM), which can implement highly parallel pattern matching and distance-based similarity measure in memory, has triggered a lot of attention for data-centric machine learning (ML) applications [1-6] (Fig. 1a). However, conventional SRAM-based CAM cell requires at least ten transistors [7], limiting the density increase for compute-intensive algorithm mapping. Although various kinds of emerging non-volatile memories (NVMs), such as RRAM, PCM and FeFET, have been applied to construct the CAM cells with the lower hardware cost [4-6], the implementations are all based on twin complementary circuit branch path topology to represent the XNOR-like matching operations of CAM with at least two entry state (Fig. 1bc). Moreover, by utilizing the multi-level NVMs, multi-bit CAM (MCAM) design is considered to further boost CAM density, which generally requires complex writing operation for complementary multi-bit storages and extra circuit like linear analog inverter for complementary multi-bit searching signal of twin branches [5-6], suffering from extra hardware cost and energy consumption.

In this work, by utilizing the unique ambipolar effect, a novel ferroelectric tunnel FinFET (Fe-FinTFET) based CAM with only one transistor is proposed and experimentally demonstrated for the first time. Without the need of complementary branches for XNOR-like logic, the single transistor based cell also enables multi-bit CAM realization and both NOR- and NAND-type array macro design. Based on the design, Hamming and Manhattan distance computing are demonstrated for ML applications with high energy efficiency.

Experiments of Fe-FinTFET for CAM

The FinTFET structure of this work is experimentally fabricated based on 14-nm FinFET technology node (Fig. 2a). The device layout without drain underlap region is designed to realize the distinct ambipolar band-to-band tunneling (BTBT) current of tunnel FET (TFET). The proposed Fe-FinTFET is constructed by connecting a 5nm Hf_{0.5}Zr_{0.5}O₂ ferroelectric (FE) layer to the gate of FinTFET. When applying relatively large V_G , the I_D - V_G curves of fabricated Fe-FinTFET show ferroelectric hysteresis behavior with ambipolarity (Fig. 2c) due to the nonvolatile FE polarization switching behavior. Moreover, the modulated channel conductance of Fe-FinTFET can be retained after large programming voltage (V_P) pulses applied, indicating the storage of entry (Fig. 3a). Additionally, with continuous V_P pulse applied, the channel conductance modulation is non-monotonic which is mainly due to the varied tunneling current from drain junction to source junction (Fig. 3b). After programming, the I_D - V_G curves of fabricated Fe-FinTFET under relatively small V_G range show non-destructive behavior with negligible hysteresis (Fig. 3c).

Fe-FinTFET-based Ternary CAM

The stored entry of CAM cell based on single ferroelectric TFET

(FeTFET) transistor is represented by the different non-volatile FE polarization state, which can modulate the device threshold voltage and thus the V_G at the minimum I_D level (V_{OFF}) (Fig. 4b). In searching phase, V_G with limited amplitude is applied for different input searching query of CAM (V_{SL}). As shown in the truth table (Fig. 4c), due to the ambipolar BTBT current feature, the in memory XNOR logic of 1-bit entry and query ("0"/"1" state) can be carried out in one FeTFET, and only when V_{SL} is equal to the stored V_{OFF} state, can the device be turned off and thus the match-line discharge current (I_{ML}) is relatively low. For ternary CAM (TCAM), the "don't care" entry ("X" state) can be implemented by modulating FE state to ensure the V_{OFF} in the middle of "0" and "1" states, by exploiting the low subthreshold current feature irrespective of input searching query, as shown in the searching operation sequences based on FeTFET cell (Fig. 4d). For our fabricated Fe-FinTFET device, as shown in Fig. 5, after the entry of "1", "0" or "X" is programmed into Fe-FinTFET with different V_P , the searching voltage pulse are applied to the gate of the device, where $V_{SL}=1V$ represents searching "1" and $V_{SL}=-1V$ represents searching "0". Only in the case of mismatch between stored data and searching data, can the drain current I_{ML} of Fe-FinTFET be sufficiently high for discharging, indicating the XNOR-like compare operation of TCAM. Moreover, the discharging speed of comparing entry vector and query vector is positively correlated with the number of mismatching bits (Fig. 5cd), indicating its potential for Hamming distance computing.

Fe-FinTFET-based Multi-bit CAM

By further utilizing the multi-level ferroelectric polarization states, the V_{OFF} state of FeTFET can be continuously modulated (Fig. 6a), which can realize multi-bit entry cell for CAM. As shown in the Fig. 6b, only when the multi-level V_{SL} and stored V_{OFF} state matches, can the voltage of pre-charged ML (V_{ML}) keeps high in the searching phase. As shown in Fig. 6c of measured results, the V_{OFF} is programmed to four different states respectively by various V_P in Fe-FinTFET, realizing 2-bit data storage. In the searching phase, V_{SL} corresponding to 2-bit query is applied. Only when matched condition, the device is fully turned off and the I_{ML} is ultra-low due to the ultra-low OFF-state leakage current of TFET device. The I_{ML} is positively correlated with the degree of mismatch for Fe-FinTFET MCAM, which can be used for computing absolute difference like Manhattan distance.

Furthermore, the ambipolar I_D - V_G characteristics also enable the single FeTFET device as analog CAM (ACAM) for non-overlapping intervals of continuous analog levels matching (Fig. 7). The advantage of steep subthreshold swing (SS) in TFET can also result in the more non-overlapping intervals without sensing margin penalty as shown in Fig. 7a, indicating the even higher storage density.

FeTFET-based CAM Macro

Different from conventional twin complementary branches based CAM cell, since the FeTFET CAM cell only needs one transistor, both NOR and NAND type CAM macro can be implemented for speed-dominated or power-dominated applications (Fig. 8). Based on the proposed FeTFET CAM, the computation of Hamming distance and Manhattan distance widely used for ML algorithms are demonstrated with significant improvement in speed and energy (2500X and 115X) over GPU (Fig. 9).

Summary

This work reports the first experimental demonstration of Fe-FinTFET-based CAM cell with only one transistor without conventional twin complementary circuit, enabling XNOR-like matching operation of both ternary and multi-bit CAM, and supporting NOR- and NAND-type array design (Table 1). Based on the proposed design, Hamming and Manhattan distance computing are demonstrated with high energy efficiency, showing its great potential for area- and energy-efficient machine learning application.

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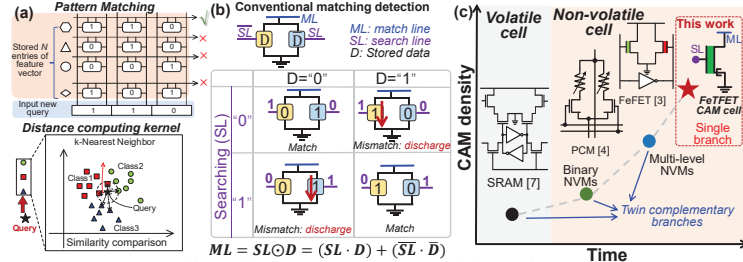


Fig. 1 (a) Content addressable memory (CAM) is widely used for in-memory pattern matching and distance computing of machine learning application. (b) Conventional CAM cell based on twin complementary branches for XNOR-like matching operation. (c) The proposed ferroelectric tunnel FET (FeTFET) based single branch CAM cell for further improved CAM density.

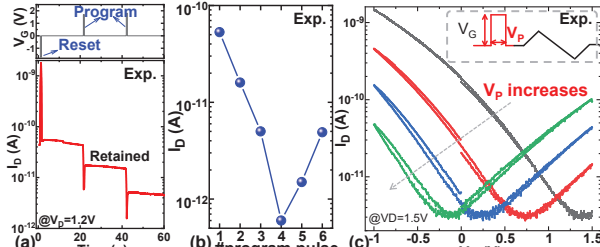


Fig. 3 Experimental characterization of Fe-FinTFET. (a) The channel conductance modulation with nonvolatility and (b) non-monotonicity due to ambipolarity. (c) I_D - V_G curves with relatively small gate voltage sweep, showing non-destructive behavior and ambipolar current effect.

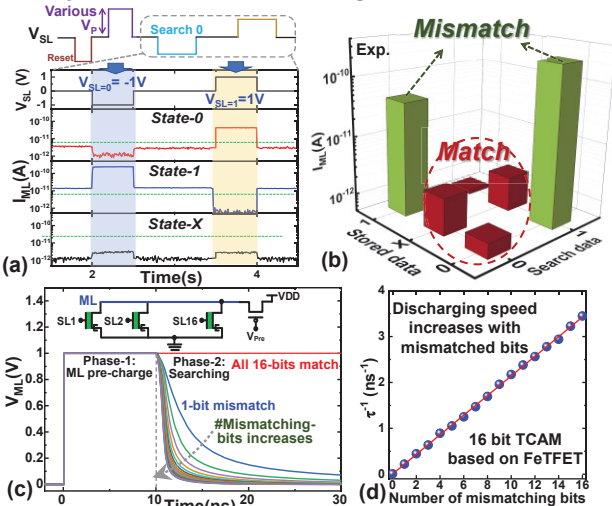


Fig. 5 Measured (a) searching operation of Fe-FinTFET-based TCAM. (b) The ML discharging current (I_{ML}) shows XNOR-like logic for matching detection of TCAM. (c) The ML discharging process of TCAM cell vector and (d) extracted time constant, which shows linearity with the number of mismatched bits.

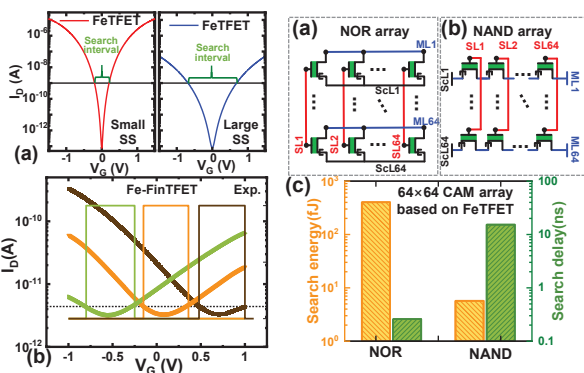


Fig. 7 (a) Analog CAM (ACAM) design based on FeTFET. (b) Measured ACAM behavior. (c) NOR- and (b) NAND-type CAM macro design and (c) comparison of energy and speed.

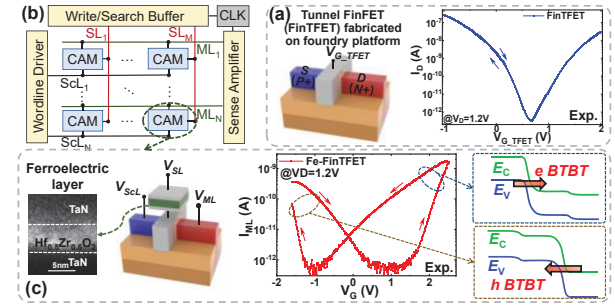


Fig. 2 Measured I_D - V_G curves of (a) underlying TFET. (b) The designed Fe-FinTFET for CAM. (c) Ambipolar behavior of Fe-FinTFET due to band-to-band tunnel (BTBT) in source/drain junction.

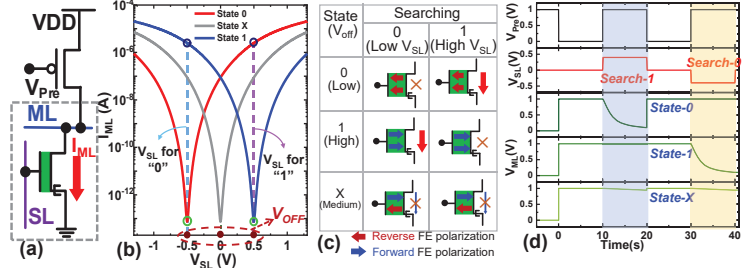


Fig. 4 Design of proposed FeTFET for ternary CAM (TCAM). (a) Simplified matching detection schematic diagram. (b) Transfer characteristics of FeTFET with various writing states based on calibrated TFET model [8] and (c) the truth table of CAM cell. (d) Sequence diagram of searching process of TCAM.

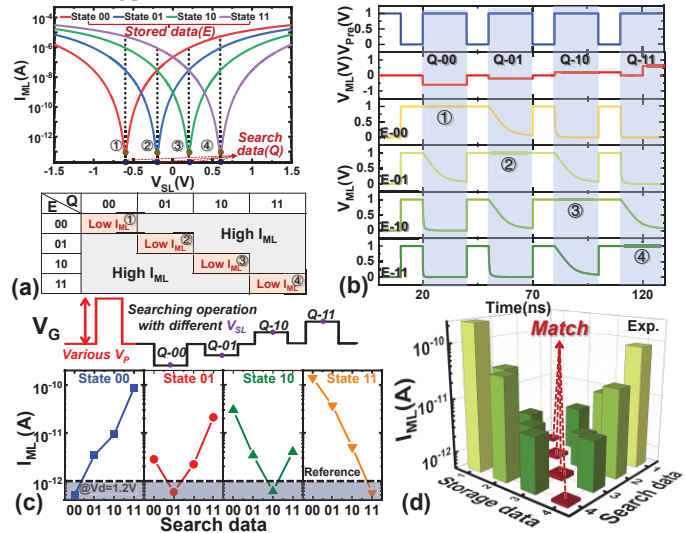


Fig. 6 Design of proposed FeTFET for multibit CAM (MCAM). (a) FeTFET transfer curves with multi-level FE polarization states and truth table. (c) Matching detection of MCAM cell. Measured (c) searching operation of FeFinTFET-based MCAM and (d) ML discharging current.

Table. I Benchmark of the proposed FeTFET based CAM with other reported works.

Reference	[7]	Ref.[4]	Ref.[6]	Ref.[2-3]	This work
CAM cell	16T-SRAM	2T-2PCM	6T-2RRAM	2FeFET	1Fe-FinTFET
Multi-bit cell	No	No	Yes	w/Linear INV	Yes
Cell size	16T	2T+2R	6T+2R	>2T	1T
Topology	Complementary twin branches				Single branch
Transistor feature	Conventional				Ambipolarity
Write operations	2 times per cell				One per cell
Search energy	4.01pJ	2.62pJ	~	1.6pJ	0.41pJ

Fig. 9 Computing for (a) Hamming distance with TCAM mode and (b) Manhattan distance with MCAM mode. (c) Performance comparison.